

MTWARA, Tanzania, United Republic of

WMO: 639710

Lat: 10.339S

Lon: 40.182E

Elev: 371

StdP: 14.50

Time Zone: 3.00 (E03)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	64.9	66.1	58.1	73.1	79.2	60.3	79.3	78.2	26.3	79.4	24.4	77.8	6.9	210	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	12.4	90.0	77.3	89.5	77.4	88.2	77.2	80.7	86.3	80.0	85.6	79.5	85.0	15.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	153.3	83.7	78.8	151.5	83.4	78.2	148.3	82.9	44.8	86.5	44.1	85.6	43.4	85.1	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	22.1	20.7	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.9	81.2	80.9	80.7	79.7	78.4	76.7	75.8	75.9	76.6	78.4	80.8	81.7
	DBStd	2.67	1.97	1.71	1.76	1.72	1.36	1.40	1.32	1.63	1.64	1.81	1.51	1.70
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	10548	967	866	953	891	882	802	801	803	798	881	923	981
	CDD65	5073	502	446	488	441	417	352	336	338	348	416	473	516
	CDH74	43022	5173	4368	4385	3360	2897	2233	2142	2257	2610	3506	4660	5431
	CDH80	13478	1505	1293	1413	966	836	619	555	645	831	1230	1742	1843
Wind	WSAvg	11.0	11.4	10.6	8.7	10.9	13.4	13.5	14.0	11.7	10.1	9.8	8.9	9.4
Precipitation	PrecAvg	45.9	8.3	8.1	9.1	7.1	2.0	0.5	0.5	0.4	0.5	1.3	2.2	6.3
	PrecMax	66.3	22.0	39.4	29.8	15.3	7.2	3.4	2.4	2.2	1.9	6.5	8.9	16.8
	PrecMin	30.2	0.9	1.3	2.8	1.9	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.5
	PrecStd	8.1	5.1	6.0	4.5	3.1	1.6	0.6	0.6	0.5	0.5	1.5	2.2	4.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.7	89.7	90.4	89.7	88.0	87.8	87.1	88.0	89.6	91.2	91.4	91.3
		MCWB	79.0	79.3	79.4	78.5	76.0	72.9	71.7	71.5	72.8	75.1	75.9	78.1
	2%	DB	88.2	88.2	89.4	88.1	86.4	86.2	85.8	86.3	87.7	89.0	89.8	89.7
		MCWB	78.8	79.0	79.0	78.2	75.6	72.7	71.2	71.4	72.6	74.9	76.3	78.0
	5%	DB	87.6	87.6	88.1	87.2	85.9	84.7	84.4	85.5	86.1	87.4	88.9	88.4
		MCWB	78.6	78.9	78.8	77.9	75.4	72.2	71.1	71.4	72.4	74.7	76.3	78.1
	10%	DB	86.2	86.3	87.1	85.9	84.6	83.9	83.3	84.0	84.6	86.0	87.8	87.7
		MCWB	78.4	78.6	78.7	77.5	75.0	71.9	70.9	71.5	72.5	74.6	76.1	78.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.9	81.1	81.7	80.7	78.8	76.5	76.9	76.4	76.9	78.1	79.2	81.0
		MCDB	85.7	86.1	87.4	86.5	84.7	83.5	83.0	84.5	83.8	85.9	86.2	86.6
	2%	WB	80.2	80.1	80.7	79.6	77.8	75.3	74.0	74.6	75.3	77.0	78.5	80.1
		MCDB	85.3	85.4	86.4	85.2	83.8	82.4	82.0	82.5	83.2	85.0	85.5	86.0
	5%	WB	79.5	79.5	79.9	79.0	77.1	74.5	72.9	73.5	74.4	76.2	77.9	79.4
		MCDB	84.5	84.7	85.5	84.5	83.0	81.6	81.0	81.4	82.4	84.0	84.9	85.0
	10%	WB	79.0	79.0	79.1	78.4	76.3	73.6	72.1	72.6	73.6	75.4	77.2	78.9
		MCDB	83.9	84.1	84.7	83.6	82.1	80.8	80.1	80.4	81.7	83.2	84.3	84.5
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.6	13.2	12.6	14.1	16.1	17.2	17.9	18.3	16.9	15.5	12.4
		MCDBR	11.7	13.2	14.7	14.4	15.2	16.9	18.0	19.2	19.7	18.3	16.5	14.0
		MCWBR	5.0	5.7	6.7	6.4	6.2	6.6	6.7	7.0	7.4	7.0	5.7	5.4
	5% WB	MCDBR	10.9	11.9	13.6	13.1	13.8	15.9	17.3	17.9	17.8	16.9	15.1	12.3
		MCWBR	5.3	5.9	6.9	6.5	6.5	7.1	7.5	8.1	8.0	7.1	6.0	5.5
Clear-Sky Solar Irradiance	taub	0.466	0.451	0.411	0.385	0.381	0.370	0.372	0.381	0.395	0.409	0.412	0.425	
	taud	2.311	2.372	2.501	2.574	2.556	2.561	2.526	2.484	2.434	2.418	2.432	2.439	
	Ebn at Noon	279	282	289	287	279	277	279	284	289	291	293	291	
	Edh at Noon	44	41	36	31	30	30	31	34	37	39	39	39	
All-Sky Solar Radiation	RadAvg	1715	1717	1677	1633	1597	1531	1569	1730	1893	1990	2037	1883	
	RadStd	172	122	119	111	79	51	52	41	52	63	81	132	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page